

Summary Report

AnalystLab Africa Data Analytics Internship Program

Week 2: Data Cleaning & Exploratory Data Analysis (EDA)

Dataset: Netflix Movies & TV Shows Dataset

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Cleaning Challenges Encountered

During the data cleaning process, several challenges were identified. Missing values were present in the **director**, **cast**, **country**, **date_added**, **rating**, and **duration** columns. To preserve valuable information, missing values in the **director**, **cast**, and **country** columns were replaced with "Unknown", while records with missing values in **date_added**, **rating**, and **duration** were removed because they represented a very small proportion of the dataset and were essential for analysis.

Another challenge occurred while converting the **date_added** column from an object data type to a datetime format. The conversion initially failed because some records contained leading spaces. This issue was resolved by removing the extra spaces using the `str.strip()` method before applying `pd.to_datetime()`. The dataset was also checked for duplicate records, and none were found. Finally, column names and data types were standardized to ensure consistency throughout the dataset.

Key EDA Findings

The exploratory data analysis showed that **Movies** dominate Netflix's content library with **6,126 titles**, compared to **2,664 TV Shows**. Content additions increased steadily from **2016**, reaching their highest level in **2019** before declining slightly in subsequent years.

The **United States** was identified as the largest content-producing country, followed by **India** and the **United Kingdom**. The analysis also revealed that **TV-MA** is the most common audience rating, indicating that much of Netflix's content is targeted toward mature viewers. Furthermore, **International Movies**, **Dramas**, and **Comedies** were the most frequently occurring genres in the dataset, reflecting Netflix's emphasis on diverse and globally distributed content.

Top Insights

- Movies represent approximately **70%** of Netflix's content library, making them the platform's dominant content type.
- Netflix experienced rapid growth in content additions between **2017 and 2020**, with **2019** recording the highest number of titles added.
- The **United States** remains Netflix's largest content contributor, while India and the United Kingdom are also major production markets.
- The prevalence of **TV-MA** and **TV-14** ratings suggests that Netflix primarily serves mature and teenage audiences.
- The popularity of **International Movies**, **Dramas**, and **Comedies** demonstrates Netflix's strong investment in diverse global entertainment.